

UrbanMobility Transit Analytics

Checkpoint 2 · Week 3

Data Pipeline + Baseline Metrics

This checkpoint focuses on developing a repeatable data pipeline, establishing baseline metrics, and maintaining version control for the analysis workflow. The objective is to create a consistent foundation for analyzing transit usage, passenger demand, travel duration, route performance, and peak-hour activity.

1. Data Pipeline

A repeatable data pipeline was designed using Python. The workflow begins by loading the raw transit dataset and checking its structure and data quality. Missing values, duplicate records, inconsistent formats, and invalid entries are identified and handled. The cleaned data is then transformed into a standardized format suitable for analysis.

The pipeline is organized into reusable steps: data loading, data-quality checking, cleaning, transformation, feature preparation, metric calculation, and output generation. This approach makes the workflow easier to rerun when new transit records become available.

2. Baseline Metrics

Metric	Purpose
Total Trips	Measures overall transit usage.
Total Passenger Volume	Shows the number of passengers served.
Average Trip Duration	Provides a baseline for travel time.
Average Distance	Measures typical journey length when distance is available.
Peak-Hour Demand	Identifies periods with higher passenger/trip activity.
Route Performance	Compares trip activity and travel patterns across routes.

3. Data Quality Checks

The dataset was reviewed for missing values, duplicate records, inconsistent date and time formats, incorrect data types, and unusual or invalid values. These checks help improve the reliability of the calculated metrics and reduce the risk of incorrect conclusions during later analysis.

4. Version Control

Version control was established for the analysis scripts using Git and GitHub. The repository is structured to keep raw data, analysis notebooks, reusable Python scripts, generated outputs, and project documentation organized. Changes to scripts can be tracked through commits, making the analysis reproducible and easier to maintain.

5. Repository Structure

Folder/File	Description
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data/	Raw or prepared transit dataset
scripts/data_pipeline.py	Reusable data cleaning and metric pipeline
notebooks/	Exploratory analysis notebook
outputs/	Baseline metrics and processed results
README.md	Project overview and instructions
requirements.txt	Python packages required for the analysis

6. Work Summary

For the UrbanMobility Transit Analytics project, I developed a repeatable Python-based data pipeline to collect, clean, preprocess, transform, and analyze transit data. I performed data-quality checks and calculated baseline metrics including total trips, passenger volume, average trip duration, route-wise performance, and peak-hour demand. The analysis scripts were organized and maintained using Git and GitHub for version control and reproducibility. This workflow provides a reliable foundation for future visualization, trend analysis, and data-driven decision-making.

Tools: Python · Pandas · NumPy · Jupyter Notebook · Git · GitHub